

PROJECT TITLE :COLOUR SORTING

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ABSTRACT

A color sorting machine is an automated system designed to identify and separate objects based on their color using image-based or sensor-based techniques. This project presents a low-cost color sorting mechanism that uses a TCS3200 color sensor, a microcontroller, and a conveyor system driven by servo motors. The sensor detects the RGB values of each object and classifies it into predefined color categories. According to the detected color, the machine activates specific actuators that direct the object to its corresponding bin. The developed system demonstrates efficient, accurate, and fast color-based sorting suitable for small-scale industries, educational laboratories, and automation learning.

INTRODUCTION

Automation plays a crucial role in improving manufacturing efficiency, accuracy, and productivity. Color sorting is widely used in food processing, recycling, packaging, and quality control systems. Traditionally, human operators performed manual sorting, which is slow, inconsistent, and prone to error.

The color sorting machine utilizes a color sensor and microcontroller-based control logic to automate the sorting process. This eliminates human error, reduces labor costs, and maintains high sorting consistency. The system is simple, cost-effective, and easy to replicate for educational or industrial automation purposes.

AIM

To design and develop a microcontroller-based color sorting machine that identifies objects by color and automatically sorts them into appropriate categories using actuators.

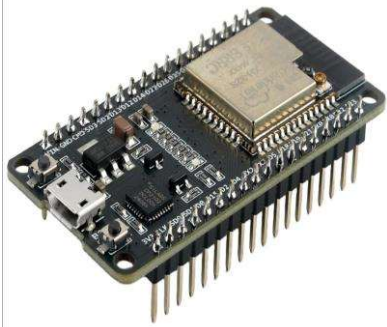
COMPONENTS

1. ESP-32
2. TCS3200 Color Sensor
3. Servo Motors (SG90)
4. IR Sensor
5. Conveyor belt mechanism (DC motor/roller system)
6. Motor Driver Module (L293D)
7. 5V Power Supply
8. Connecting wires, breadboard
9. Frame/Structure (wood)

SPECIFICATIONS OF COMPONENTS

1. ESP32

- Microcontroller: ATmega328P
- Operating voltage: 3.3V
- Clock Speed: 240 MHz
- Digital I/O pins: 14
- Analog pins: 18



2. TCS3200 Color Sensor

- Output: Square wave frequency proportional to color intensity
- Detection range: Red, Green, Blue
- Operating voltage: 3–5V
- High-resolution sensing



3. Servo Motor (SG90)

- Operating voltage: 4.8–6V
- Rotation: 0–180°
- Torque: 1.8–10 kg-cm depending on model



4. DC Motor (for conveyor)

- Voltage: 6–12V
- Speed: 100–600 RPM
- Controlled via motor driver

5. Motor Driver (L298N)

- Operating voltage: 5–35V
- Dual full H-bridge
- Peak current: 2A

6. Power Supply

- 5V for logic and servo
- 12V (optional) for conveyor DC motor

WORKING PRINCIPLE

1. Object Detection:

An object placed on the conveyor belt approaches the color sensing area. An IR sensor used to detect the presence of the object.

2. **Color Sensing:**

The TCS3200 sensor emits light and measures the reflected RGB frequency values from the object.

3. **Color Processing:**

The Arduino reads the RGB values and compares them with predefined thresholds to classify colors (e.g., red, green, blue).

4. **Sorting Mechanism:**

Based on the detected color Servo motor rotates to direct the object into the correct bin.

5. **Object Collection:**

The object is pushed into the corresponding color container.

6. **Continuous Operation:**

The conveyor keeps moving, allowing multiple objects to be sorted sequentially.

CIRCUIT DIAGRAM (TEXT DESCRIPTION)

- **TCS3200 sensor** connected to Arduino pins:

- S0 → D25
- S1 → D33
- S2 → D32
- S3 → D27
- OUT → D34

- **Servo motor** connected to Arduino:

- Signal → D23
- VCC → 5V
- GND → GND

- **DC motor (conveyor) through L298N:**
 - Motor outputs → Conveyor motor
 - IN1, IN2 → D2, D3
 - 12V supply → L298N
 - **IR Object Sensor:**
 - OUT → D4
 - VCC → 5V
 - GND → GND
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- **Common Ground** shared by all components.

RESULT

The developed prototype successfully detects and classifies objects based on color. The system accurately identifies red, green, and blue objects with a high success rate under controlled lighting conditions. The sorting mechanism responds quickly to sensor outputs and directs the objects to the correct bins without collision or misplacement. This demonstrates the feasibility of using low-cost components to build an effective automated sorting machine.

CONCLUSION

The color sorting machine achieves automated classification of objects using a TCS3200 sensor and microcontroller-based logic. It significantly reduces manual effort and increases sorting accuracy. The system is scalable and can be enhanced with machine vision, improved mechanical design, and support for multiple colors. The project provides a strong foundation for learning automation, robotics, sensors, and industrial sorting systems.

APPLICATIONS

1. **Food processing industries** – sorting seeds, grains, pulses.
2. **Recycling plants** – separating plastic based on color.
3. **Pharmaceutical companies** – capsule sorting.
4. **Quality control in manufacturing** – detecting defective color-coded parts.
5. **Packaging industries** – color-based product categorization.
6. **Educational purposes** – robotics and automation learning projects.

REFERENCES

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5. Research papers on color sorting mechanisms and automation techniques.